Module 04 Lab - Exploratory Data Analysis**Objective:** To learn how to explore a dataset to find patterns, anomalies, and insights before modeling. EDA is like being a detective; you are looking for clues in the data that will help you build a better model.**This lab is fully coded.** Your task is to run each cell, read the detailed explanations, understand the purpose of each visualization, and then complete the experimentation section.

Part 1: Setup and Data Loading**What is Exploratory Data Analysis (EDA)?**EDA is the process of using summary statistics and visualizations to understand a dataset's main characteristics. Before you can build a model, you need to understand your data. What stories does it tell? Are there errors or missing values? Are there strong relationships between variables? EDA helps answer these questions.We will use the famous Titanic dataset for this lab. It contains information about passengers and, crucially, whether they survived the disaster.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Seaborn is a library built on top of Matplotlib that makes creating beauti1

# Load the dataset directly from a URL
df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/

print("--- First 5 Rows ---")
print(df.head())
```

```
print("--- Basic Info ---")
# .info() is a great first command. It tells us the column names, how many nc
# Notice that 'Age' and 'Cabin' have missing values!
df.info()
```

```
--- First 5 Rows ---
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                 Name     Sex   Age  SibSp  \
0                             Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                              Heikkinen, Miss. Laina  female  26.0      0
3        Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                            Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
--- Basic Info ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

Part 2: Descriptive StatisticsLet's start by getting a high-level numerical summary of the data. The `.describe()` method is perfect for this. It calculates statistics like mean, standard deviation, min, and max for the numerical columns.

```
# Get summary statistics for numerical columns
print("--- Descriptive Statistics ---")
print(df.describe())

print("--- Key Insights from Statistics ---")
print(f"The average age of a passenger was {df['Age'].mean():.1f} years.")
print(f"The overall survival rate was {df['Survived'].mean():.1%}.")
print(f"Fares ranged from ${df['Fare'].min():.2f} to a whopping ${df['Fare'].
```

```
--- Descriptive Statistics ---
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
--- Key Insights from Statistics ---
The average age of a passenger was 29.7 years.
The overall survival rate was 38.4%.
Fares ranged from $0.00 to a whopping $512.33.
```

Part 3: Visual EDA - Telling Stories with PlotsNumbers are great, but plots make patterns and relationships immediately obvious. The goal of visual EDA is to turn data into insights.**A Note on Plotting Libraries:*** **Matplotlib:** The foundational library, gives you full control over everything.* **Seaborn:** Built on Matplotlib, it provides a simpler, high-level interface for creating common statistical plots. We will use Seaborn for its ease of use and attractive defaults.

Visualization 1: How many survived?A simple `countplot` is the best way to see the distribution of a categorical variable, like our target `Survived`.

```
sns.set_style('whitegrid')  # Sets a nice visual style for our plots

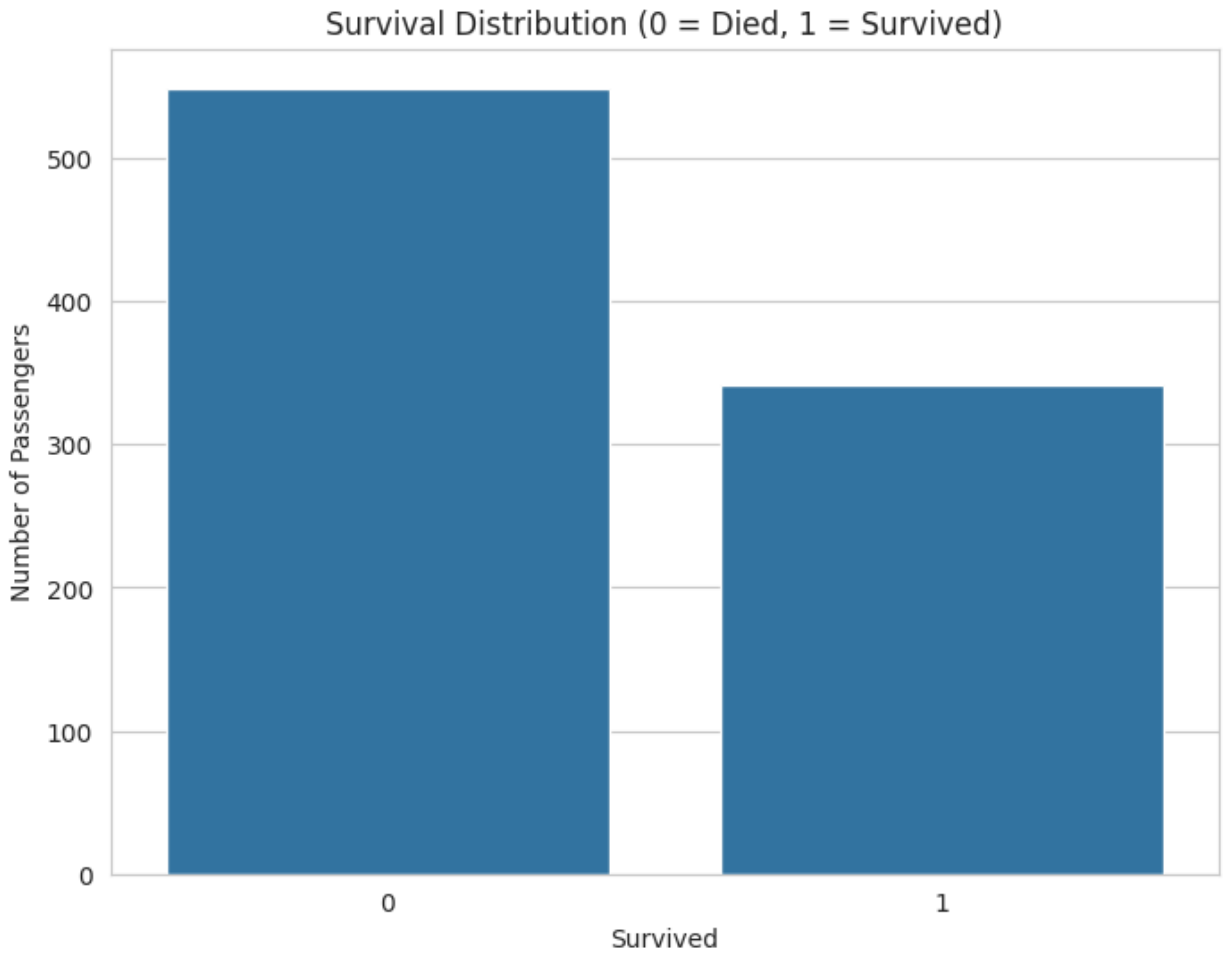
plt.figure(figsize=(8, 6))

sns.countplot(x='Survived', data=df)

plt.title('Survival Distribution (0 = Died, 1 = Survived)')
plt.xlabel('Survived')
plt.ylabel('Number of Passengers')

plt.show()

print("Insight: Far more people died than survived. This shows that the datas
```




```
Insight: Far more people died than survived. This shows that the dataset is sc
```

Visualization 2: Does passenger class matter for survival?Now we want to see if there's a relationship between two variables. We can use the `hue` parameter in `countplot` to split the bars by another category.

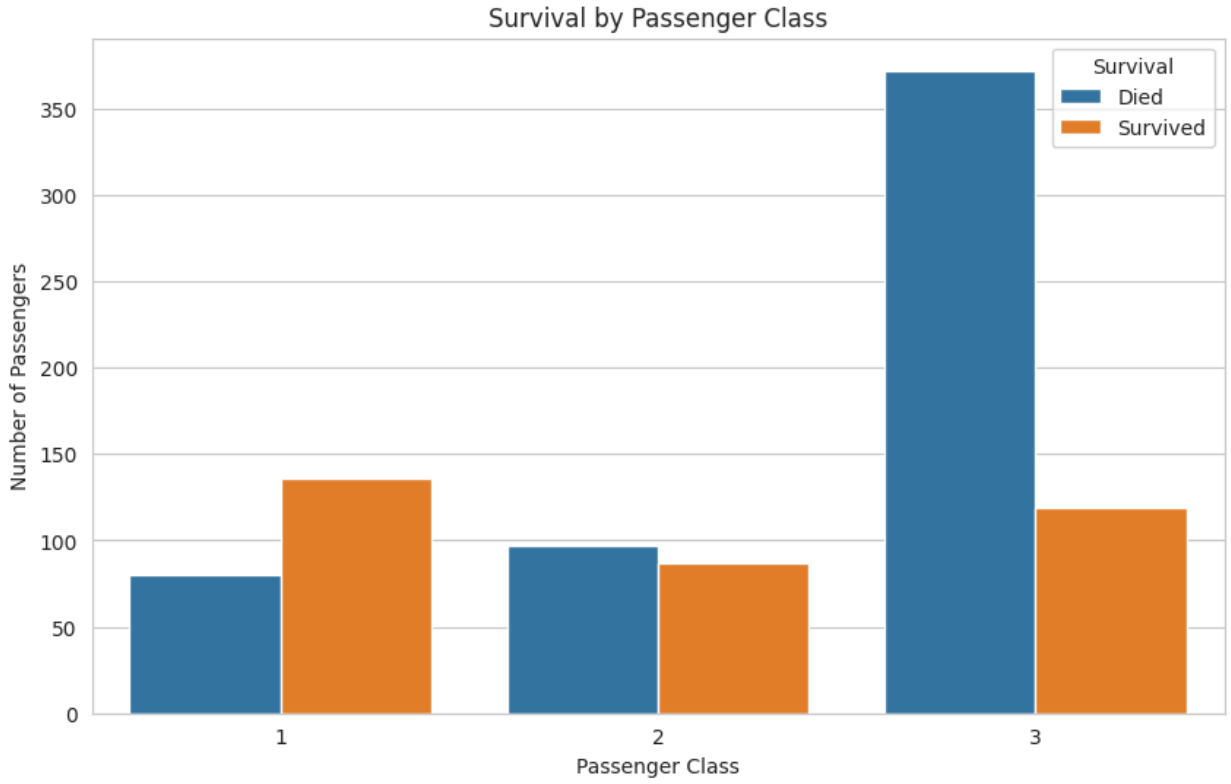
```
plt.figure(figsize=(10, 6))

sns.countplot(x='Pclass', hue='Survived', data=df)

plt.title('Survival by Passenger Class')
plt.xlabel('Passenger Class')
plt.ylabel('Number of Passengers')
plt.legend(title='Survival', labels=['Died', 'Survived'])

plt.show()

print("Insight: There is a strong relationship between passenger class and su
```




```
Insight: There is a strong relationship between passenger class and survival.
```

Visualization 3: What about gender?The 'women and children first' mantra is famous. Let's see if the data supports it.

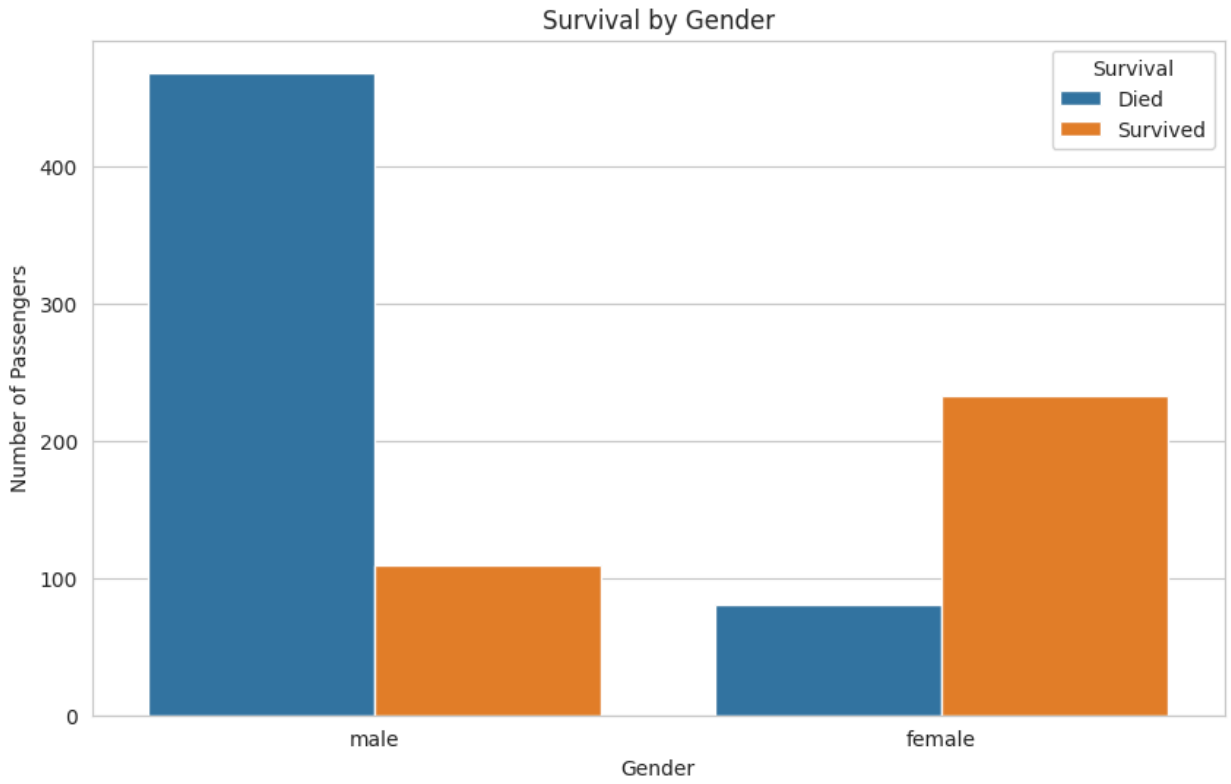
```
plt.figure(figsize=(10, 6))

sns.countplot(x='Sex', hue='Survived', data=df)
```

```
plt.title('Survival by Gender')
plt.xlabel('Gender')
plt.ylabel('Number of Passengers')
plt.legend(title='Survival', labels=['Died', 'Survived'])

plt.show()

print("Insight: Female passengers had a much higher survival rate than male p
```




```
Insight: Female passengers had a much higher survival rate than male passenger
```

Visualization 4: How does age play a role?For a continuous variable like `Age`, a histogram is a great way to see its distribution.

```
# A FacetGrid allows us to create multiple plots side-by-side
# to compare distributions.

# Create one histogram for passengers who died (Survived = 0)
# and another for passengers who survived (Survived = 1).

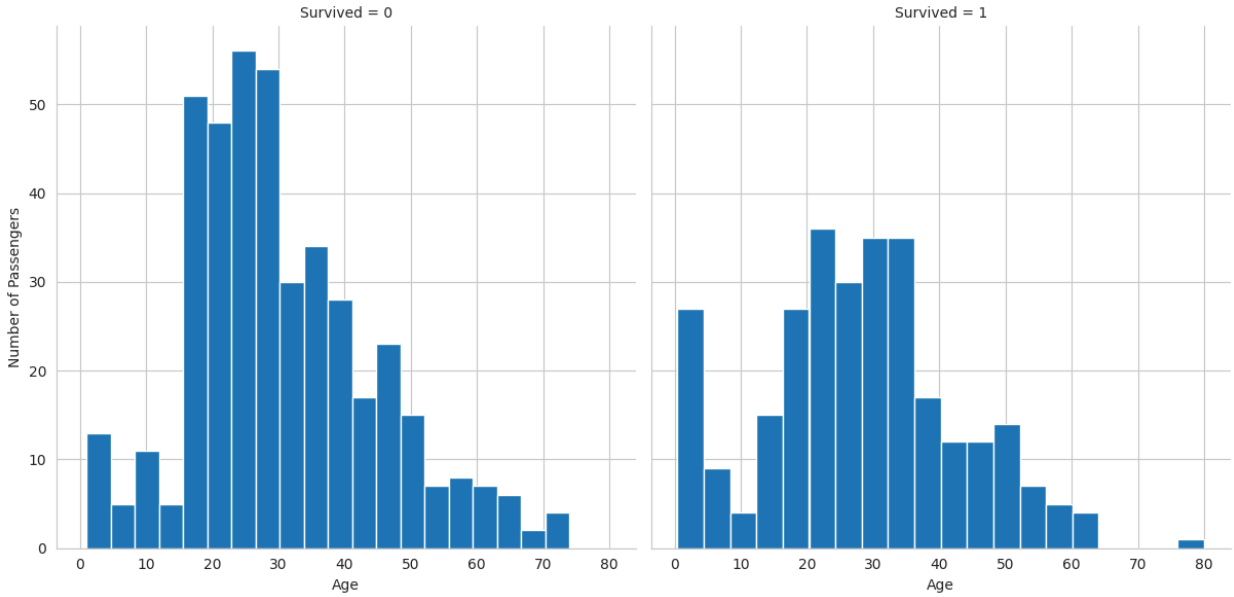
g = sns.FacetGrid(df, col='Survived', height=6)
```

```
g.map(plt.hist, 'Age', bins=20)

g.set_axis_labels('Age', 'Number of Passengers')
g.set_titles('Survived = {col_name}')

plt.show()

print("Insight: The age distribution shows that many passengers who did not s
```




```
Insight: The age distribution shows that many passengers who did not survive w
```

Part 4: Student Experimentation**Instructions:** Create your own visualizations to explore other relationships in the data.

Experiment 1: Port of Embarkation1. The `Embarked` column tells you where a passenger boarded the ship (C = Cherbourg, Q = Queenstown, S =

Southampton).2. Create a `countplot` to see how survival rates differed by the port of embarkation. Does where they boarded seem to be related to their survival?

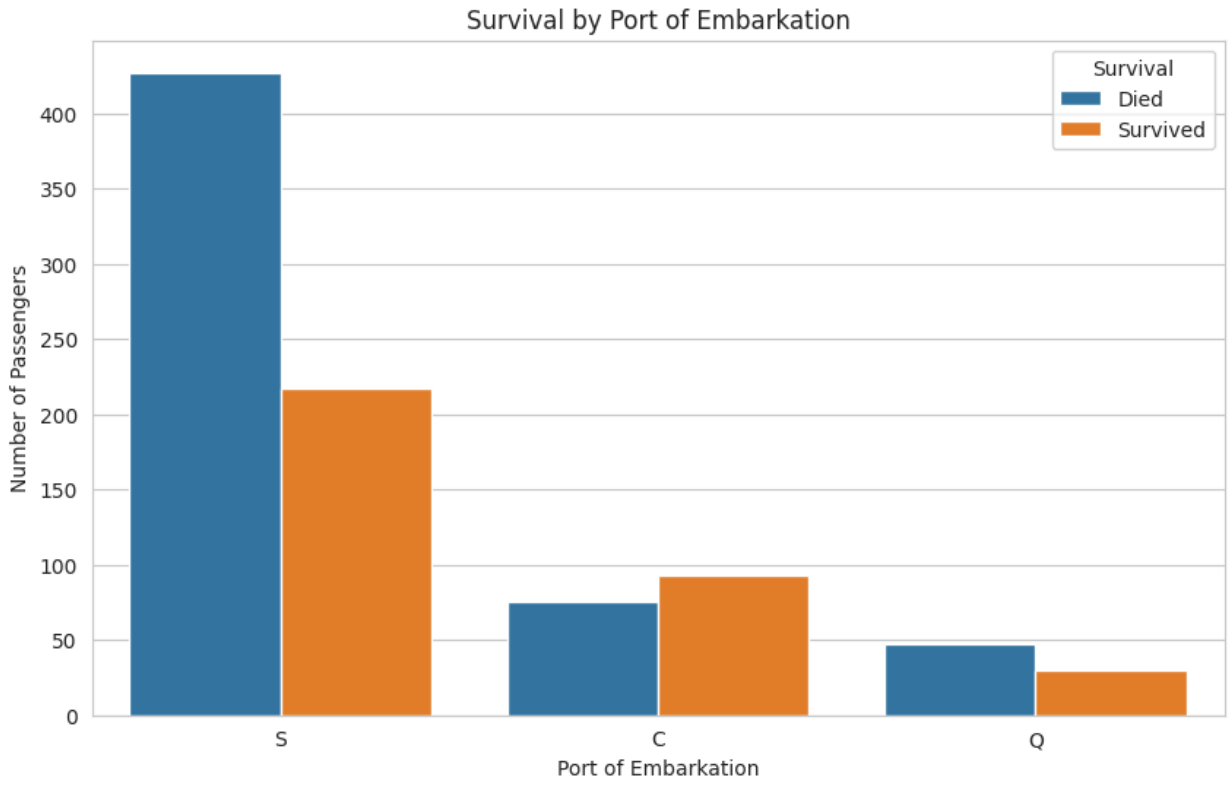
```
plt.figure(figsize=(10, 6))

sns.countplot(x='Embarked', hue='Survived', data=df)

plt.title('Survival by Port of Embarkation')
plt.xlabel('Port of Embarkation')
plt.ylabel('Number of Passengers')
plt.legend(title='Survival', labels=['Died', 'Survived'])

plt.show()

print("Insight: Survival appears to be related to the port of embarkation. Pa
```




```
Insight: Survival appears to be related to the port of embarkation. Passengers
```

Experiment 2: Fare vs. Survival1. `Fare` is a continuous numerical variable.2. A `boxplot` or `violinplot` is a great way to see the distribution of fares for those who survived vs. those who didn't.3. Create one of these plots to compare the `Fare` distribution by `Survived`. What does it tell you?

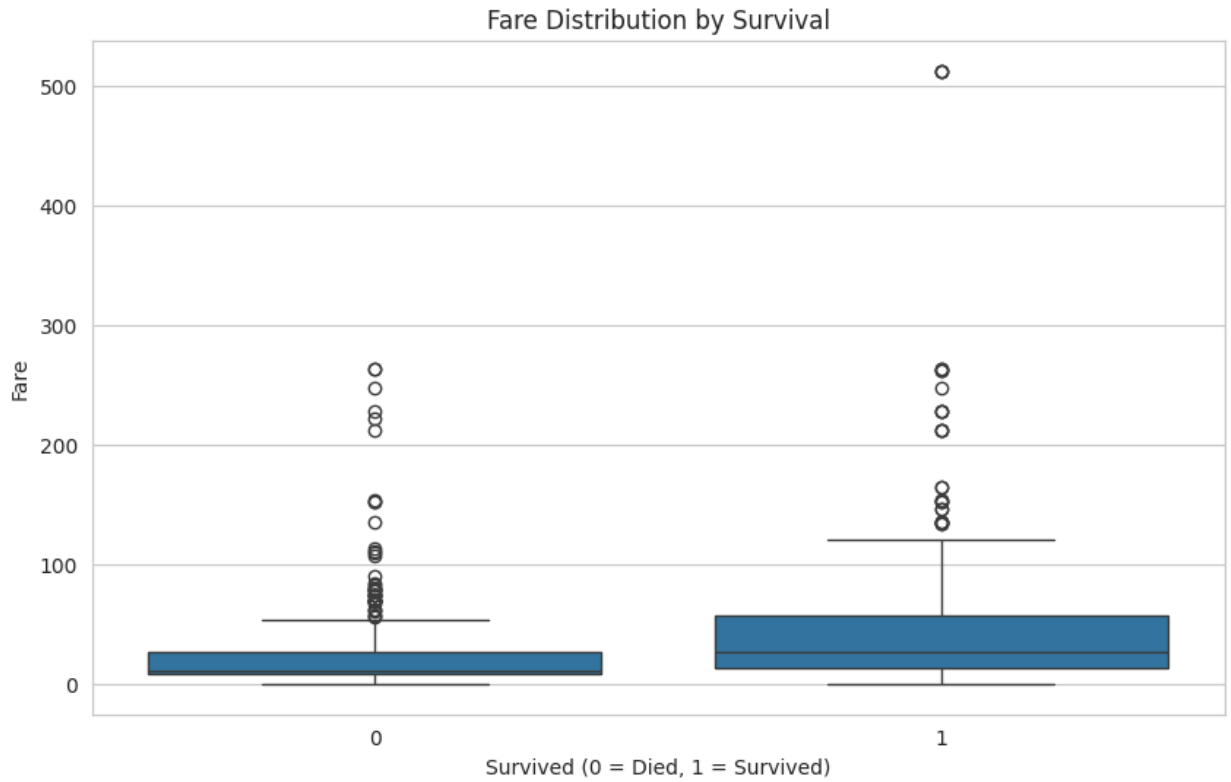
```
plt.figure(figsize=(10, 6))

sns.boxplot(x='Survived', y='Fare', data=df)

plt.title('Fare Distribution by Survival')
plt.xlabel('Survived (0 = Died, 1 = Survived)')
plt.ylabel('Fare')

plt.show()

print("Insight: Passengers who survived generally paid higher fares than pass
```




```
Insight: Passengers who survived generally paid higher fares than passengers w
```

📝 Knowledge Check**Instructions:**

1. What is the primary goal of Exploratory Data Analysis (EDA)? The main goal of EDA is to understand the data before building a machine learning model. It helps identify patterns, relationships, missing values, unusual data, and important variables that may affect the results.
2. Based on the plots in this lab, what kind of person had the best chance of surviving the Titanic? Based on the plots, a female passenger in first class, especially a younger passenger or child, had one of the best chances of surviving. First-class passengers had higher survival rates than third-class passengers, and females survived at a much higher rate than males. Age also appeared to matter, especially for children.
3. Why is it important to visualize data instead of just looking at summary statistics? Visualizing data is important because graphs can show patterns and relationships that numbers alone may not make obvious. For example, a mean can tell us the average age or fare, but a plot can show the distribution, differences between groups, outliers, and trends. Visualizations make it easier to understand the data and notice important patterns.